AI Usage Declaration

Project: Below the Ember

Student: Bekzat Zhailkhan

Course: AI Fundamentals Final Project 2026

Date prepared: 25 October 2026

Assistance used

This project used ChatGPT and Codex as development assistants. AI assistance was substantial and covered implementation suggestions and code, debugging, test creation and execution, procedural visual and audio code, algorithm explanations, documentation, presentation preparation and demo editing.

The student selected the project direction, requested iterative features and improvements, reviewed the product through the development conversation, and is the sole student participant. This declaration does not describe the code or documents as entirely unaided work.

AI inside the application

The running application uses locally implemented algorithms, including agents, A*, CSP and the additional laboratory methods. It does not call an external LLM or hosted AI API. Runtime AI behavior is separate from the AI assistance used to develop the project.

Validation and responsibility

Automated checks, browser observations and experiment outputs support the technical claims. AI output can contain mistakes; tests do not prove full correctness. The student remains responsible for reviewing the submitted files, following the course rules, understanding the algorithms, and answering questions independently during the defense.

References identify algorithm learning material and framework documentation. Third-party libraries retain their own licenses. Procedural graphics and music come from project code; the demo uses separately mixed original synthesized music and actual browser frames. No external dataset or secret AI API key is required.

Student review

This declaration is prepared for student review. No signature has been applied on the student's behalf. If the course requires a signed declaration, Bekzat Zhailkhan should review and sign it personally.

Student signature: _____

Date signed: _____